

A simple introduction to multiphoton microscopy

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Summary

Multiphoton microscopy is a powerful technique based on complex quantum mechanical effects. Thanks to the development of turnkey mode-locked laser systems, multiphoton microscopy is now available for everyone to use without extreme complexity. In this short introduction, we describe qualitatively the important concepts underlying the most commonly used type of multiphoton microscopy (two-photon excitation). We elucidate how those properties lead to the powerful results that have been achieved using this technique. As with any technique, two-photon excitation microscopy has limitations that we describe, and we provide examples of particular classes of experiments where two-photon excitation microscopy is advantageous over other approaches. Finally, we briefly describe other useful multiphoton microscopy approaches, such as three-photon excitation and second harmonic generation imaging.

Introduction

Multiphoton microscopy is a powerful technique based on nonlinear interactions between photons and matter. In this short review, we will explain in simple terms the main concepts behind this technique. It is beyond the scope of this review to describe all the current applications of multiphoton microscopy. Several extensive reviews have been published that will serve the reader for that purpose. Our goal is a simple explanation of the technique that will provide sufficient background to a novice reader to allow him/her to understand the principles that underlie the amazing results that have been achieved using multiphoton microscopy. Thus, we have avoided many fine details that are known to the experts, and instead refer the reader to more detailed descriptions of various aspects of multiphoton microscopy. Of course, there are limitations to multiphoton microscopy as well. We will summarize those with an emphasis on the conditions where

multiphoton approaches provide distinct advantages over other fluorescence microscopy approaches.

Typical fluorescence microscopy is based on a linear effect: One photon is absorbed by the fluorescent molecule, which can in turn emit a single fluorescent photon. In this case, if the excitation power is increased by a factor of two, then twice as much fluorescence is generated. By contrast, multiphoton microscopy relies on nonlinear interactions between light and matter. The most commonly used multiphoton imaging procedure is two-photon excitation microscopy, which will be the focus of this review. We will also briefly describe other nonlinear optical imaging methods, and also provide further resources for those readers who would like to learn more about those methods.

The light source

Two-photon absorption is a rare event in which two photons interact with the same molecule at the 'same' time. In practice, the same time means within a time interval less than 10^{-18} s. Under normal arc lamp illumination the probability to observe such an interaction is practically zero. Maria Göppert-Mayer predicted the possibility of two-photon absorption in 1931, but high photon fluxes in the range of 10^{20} – 10^{30} photons/(cm² s) are necessary to generate nonlinear interactions. So the first obstacle on the way to make two-photon microscopy feasible was to deliver this high photon flux to the specimen, without simultaneously vaporizing it. The fact that two-photon microscopy is a well-established technique nowadays is the result of powerful pulsed laser systems becoming available in the last decades. Since the invention of the laser a lot of experimental spectroscopy and theoretical work focused on the development and application of two-photon excitation (Masters & So, 2004). This culminated in the first demonstration of two-photon excitation microscopy in 1990, which became feasible when subpicosecond pulse mode-locked lasers were produced (Denk *et al.*, 1990). These mode-locked lasers generate ultrashort light pulses (~ 100 fs) with a repetition rate of ~ 100 MHz (that equals one pulse every 10 ns). Because the laser is essentially off 10^5 -fold more time than it is on, the average power is kept low, while providing high photon fluxes during the pulses. In other words a mode-locked

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laser compresses the laser power in small time packets so that photons are very crowded in time. Similarly, high numerical aperture objectives are used in a two-photon excitation set-up to compress the photons in space. An objective with high numerical aperture concentrates the light in a diffraction limited focal volume with a size of ~ 1 fl that is $\sim 1 \mu\text{m}^3$ (Williams *et al.*, 1994). The combination of the two effects generates an extremely high photon flux in a very small volume, so that the probability of a nonlinear excitation event is greatly increased. Because a two-photon excitation event requires two photons to interact with the same particle at the same time, the probability of an absorption event has a supralinear dependency on the excitation intensity (the probability is proportional to the square of the instantaneous laser intensity, $P \propto I^2$), which is peaked at the focal spot in the microscope. Because of the intensity-squared dependence of two-photon excitation, it sharply decreases outside the focal volume. For all these reasons in a two-photon set-up, it is possible to produce nonlinear events only in a precise region of the sample. If we combine such a light source with a scanning system like the one used in confocal microscopes, we have the ability to produce nonlinear excitation in a thin optical section of the sample, without perturbing regions of the sample outside the focal plane (Denk *et al.*, 1990; Denk *et al.*, 2006).

The signal

Although two-photon excitation can be used to cause any kind of optical transition, it is typically used to generate fluorescence. Fluorescence is the absorption and re-emission of light by a molecule. The typical fluorescence process requires a single photon of the right energy to interact directly with the molecule. The photon may be absorbed and its energy causes the molecule transition to an excited electronic state. This excited state is unstable, and in a short time (10^{-8} – 10^{-9} s) (Williams *et al.*, 1994) the molecule returns to its ground state by emitting a new photon (Fig. 1). Because all processes result in a loss of energy, the resulting emitted photon has less energy than the 'exciting' photon (alternatively, we can say that the fluorescence has a longer wavelength than the excitation light).

To generate fluorescence by a two-photon process, two photons must be absorbed simultaneously by the same molecule (Fig. 1). These photons each have just one-half of the energy that is necessary to excite the molecule. Because photon energy and wavelength are inversely related, these photons have a wavelength that is $\sim$ two times longer than the wavelength required by a single-photon excitation (e.g. a molecule requiring 380 nm excitation light can be efficiently excited by a 760 nm light, when using two-photon excitation as in Fig. 2). Because of quantum mechanical selection rules, the exact two-photon absorption spectra may vary somewhat from twice the wavelengths of the one-photon absorption spectrum. Two-photon excitation spectra

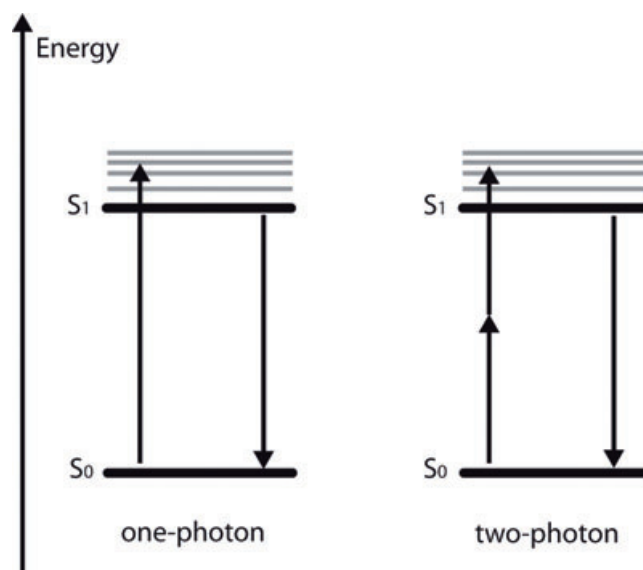


Fig. 1. Jablonski diagram illustrating one-photon versus two-photon excitation of a molecule, and the subsequent fluorescence emission.

are difficult to measure, although many have been published and are available on various websites. In practice, it is often sufficient to begin with a two-photon excitation wavelength of twice the one-photon peak absorption, and then tune the laser around that region to maximize the two-photon excitation signal. Fortunately, for biological imaging, molecules that are excited by two-photon excitation follow the same decay

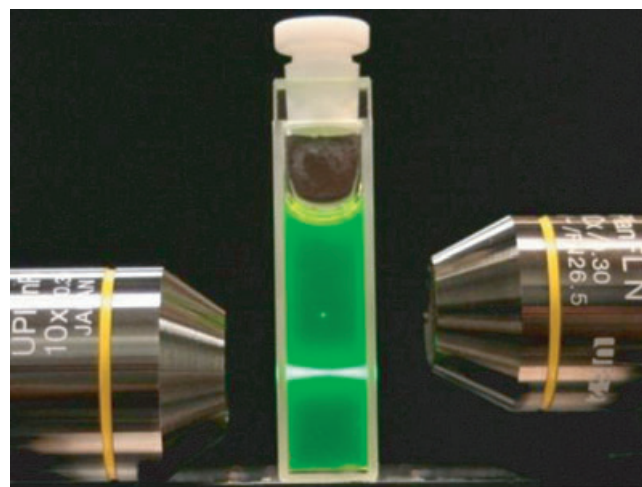


Fig. 2. Experimental illustration of the difference between linear and two-photon excitation. 380 nm (confocal) and 760 nm (two photon) excitation wavelengths were used to generate fluorescence in a 50 μM Fluorescein solution. Two-photon excitation generates fluorescence in the focal volume exclusively, whereas conventional single-photon excitation generates fluorescence all along the light path. (Image used by permission, Kevin Belfield research group, University of Central Florida; Zhen-li Huang & Ciceron Yanez)

process as for a single-photon event, and thus emit an identical characteristic fluorescence signal (Fig. 1). This leads to an apparent inverse Stokes shift, where the emitted photons have more energy than the 'exciting' photons (i.e. the fluorescence has a shorter wavelength than the excitation light).

Signal detection

The commercially available two-photon systems usually come combined to a confocal microscopy system. The scanning system of mirrors is used by both systems to scan the excitation light across the specimen, so that the image of the specimen at the focal plane is constructed. But the signal generated by two-photon excitation has different requirements for efficient detection, and also permits for a wider range of detection options. These differences are a consequence of the nonlinear nature of two-photon excitation. In a laser scanning confocal imaging system, the light being focused onto the specimen to excite the fluorescence emission interacts with fluorophores all along its path through the specimen. Therefore, we have production of fluorescence away from the focal volume (Fig. 2). A pinhole must be present in front of the detector to avoid the collection of this out-of-focus signal that would otherwise blur the image, decreasing its contrast and its spatial resolution. In this way, the pinhole leads to the much desirable optical sectioning in the confocal system, but it does so at the cost of rejecting some signal photons together with the out-of-focus ones. Further, the photons generated in the focal volume are still susceptible to scattering along their path out of the specimen. Because of this, a percentage of them changes direction, and arrive at the pinhole at a position that does not correspond to the in-focus signal. In this way, we lose a percentage of 'good' photons, and this loss becomes more significant the deeper that we image into the specimen. In fact, this is one of the key limiting factors in how deep we can image into a specimen. Various approaches to overcome this limitation have been presented in (Ntziachristos, 2010). Two-photon excitation is one of them (Helmchen & Denk, 2005).

When two-photon excitation is used, the laser light does not produce any excitation along its path through the specimen, because the photon density is sufficiently high to cause two-photon absorption only in the focal volume. This property allows more of the excitation laser beam to reach deep into the sample. Because the excitation only occurs in the focal spot of the microscope, all of the photons are generated in-focus. In other words, there is no out-of-focus signal to be rejected; on the contrary all the photons should be collected in an ideal set-up. This explains why in a two-photon excitation set-up, there is no need to have a pinhole in front of the detector. Indeed, a more simple arrangement can be used (nondescanned detection) that basically conveys all the photons collected by the objective to the photomultiplier detector. The only selection applied is based on wavelength,

by using the appropriate filter to collect the fluorescence of interest (Piston *et al.*, 1994).

One commonly asked question is how the spatial resolution of two-photon microscopy compares to confocal microscopy. With two-photon excitation, the quadratic dependence on the illumination power positively limits the size of the excitation volume, but the use of light that is double the wavelength light broadens the point-spread function. Because of its use of shorter wavelengths, an ideal confocal microscope has an intrinsic resolution half of that from a two-photon microscope (Sheppard & Gu, 1990). However, in biological fluorescence confocal microscopy, signal collection is optimized by opening the pinhole, which degrades its ideal spatial resolution. Thus, in practice two-photon microscopy provides spatial resolution that is nearly identical to confocal microscopy (Williams *et al.*, 1994).

Limitations of two-photon microscopy

Although two-photon excitation offers many advantages for laser scanning microscopy applications, it also has limitations and drawbacks. One obvious limitation is the high cost of the appropriate ultrafast laser. Although laser technology continues to improve, these lasers remain expensive (often costing as much as a complete confocal microscope). Because of this cost, it is unreasonable to have more than one two-photon excitation laser on a system, and this precludes the multiple excitation protocols that investigators rely on for multispectral confocal microscopy. A second limitation that is physical as well as practical pertains to the accelerated photobleaching in two-photon excitation. Although two-photon excitation produces photobleaching only in the focal plane, within the focal volume, high-order photobleaching is observed. In confocal microscopy, the photobleaching rate increases quasi-linearly with the excitation power, but in two-photon microscopy, the higher photon density activates more photobleaching pathways, resulting in accelerated photobleaching (Patterson & Piston, 2000; Kalies *et al.*, 2011). This high-order photobleaching is especially detrimental for thinner samples ($<10\ \mu\text{m}$), especially if the lowest possible light exposure is not used. Finally, specific limitations arise if the two-photon excitation beam interacts linearly with chromophores in the sample. For example, near infrared light is absorbed by the photosynthetic complex. Other naturally occurring chromophores, such as the pigment melanin can limit tissue imaging by causing thermal and mechanical damage during an experiment. For these reasons, two-photon excitation is often not the best method to use for a given imaging application.

Two-photon or confocal microscopy?

When is better to use two-photon microscopy and when is confocal microscopy or another imaging approach the best

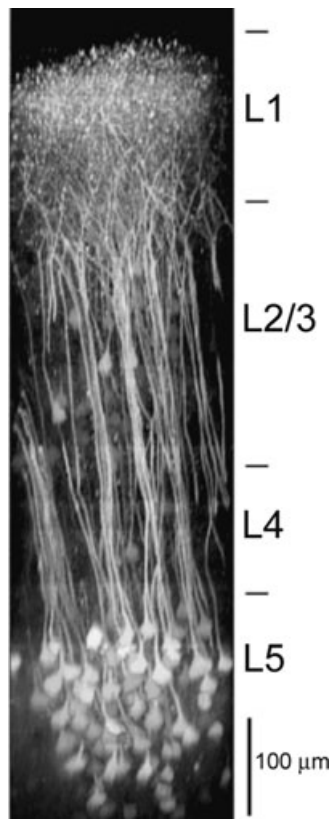


Fig. 3. Example of deep two-photon imaging in mouse neocortex. Maximum intensity side projection of a fluorescence image stack, obtained in a transgenic mouse expressing a genetically-encoded indicator. Nearly the entire depth of the neocortex can be imaged. Reprinted by permission from Macmillan Publishers Ltd: Nature Methods (Helmchen & Denk, 2005), copyright (2005).

option? This is the most common choice we must make when designing the appropriate experiment to address our scientific question. Here we list four main situations where two-photon microscopy is a valuable tool to have.

Thick specimen:

Anytime we need to image structures that extend into the specimen or that are simply buried into the specimen, optical sectioning capability is an absolute requisite. Although confocal microscopy can image at depths up to 100 μm or sometimes more into the sample, two-photon excitation can extend this limit up to 1 mm in the best conditions (Fig. 3; Ref. Helmchen & Denk, 2005). Although the ultimate imaging depth is dependent on the specific tissue under examination, a good rule of thumb is that two-photon excitation can image effectively 6-fold deeper than confocal microscopy using the same sample and fluorophores. This improvement in imaging depth arises because two-photon excitation microscopy is much less sensitive to scattering of the excitation and emission

photons, the excitation photons are not absorbed as they pass through the sample until they reach the focal spot, and two-photon excitation microscopes can use highly efficient detection strategies such as nondescanned detection. In addition, two-photon excitation microscopy uses excitation light of a longer wavelength compared to confocal microscopy. In biological tissue, longer wavelengths are less absorbed and also less affected by scattering. This allows excitation light to penetrate deeper into the specimen. Confocal microscopy is the best (and also cheaper) choice when imaging thinner specimens, unless some other factors such as those listed below need to be considered. If only a single cell layer is to be imaged, a widefield microscope is often an even better solution than confocal microscopy, because there is often no need for optical sectioning in such a thin sample.

UV excitation:

Whenever ultraviolet (UV) excitation is required to image the signal of interest from live biological samples, we have to trade out signal intensity with specimen integrity. UV photons have high energy and they can easily produce cell damage or photobleaching when high intensity is used, especially when the same region needs to be imaged repeatedly over time. More importantly, UV lasers are not commonly available with confocal systems, and when they are, special objectives are necessary to properly focus UV light. Finally, if we need to excite far from the surface of the specimen, UV photons are severely scattered and absorbed by biological tissues, thus greatly reducing the penetration of the excitation light.

None of these obstacles affect two-photon excitation microscopy at all. In fact, near infrared light can be used to excite molecules instead of requiring UV excitation, as would be needed in a confocal system. Using infrared wavelengths, the photodamage and the scattering are both reduced. So, we can use longer integration times, or increased number of scans in a time-lapse experiment without incurring in the amount of photodamage or photobleaching that UV light would have produced. Moreover, mode-locked lasers used in two-photon microscopy can generate a continuous spectrum of wavelength between 680 nm and 1000 nm, and this gives much more choices to the user, when compared to the few UV laser lines available.

In vivo imaging:

Two-photon excitation microscopy is also the best option for imaging structures *in vivo* in a living animal. First of all, it allows deeper imaging compared to confocal microscopy and this is a huge advantage, especially when excision of the specimen is not possible. Two-photon excitation microscopy also has the advantage of reduced phototoxicity, because it uses infrared light. Finally, because two-photon absorption only produces excitation in the focal plane, out-of-focus

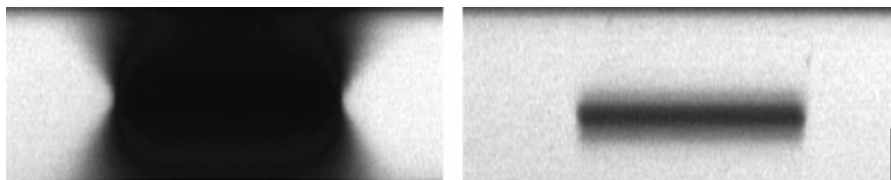


Fig. 4. Axial photobleaching patterns produced by conventional confocal and two-photon excitation after imaging a single plane inside a polymer film containing sulforhodamine B. On the left, the effect of conventional confocal illumination (514 nm); on the right, the effect of two-photon illumination (820 nm). Scale bar 5 μm .

photobleaching is greatly reduced compared to confocal microscopy, where the fluorophores are bleached evenly along the entire penetration length (Fig. 4). For all these reasons, two-photon excitation is generally the preferred method whenever *in vivo* imaging needs to be performed (Wang *et al.*, 2010).

Localized photochemistry:

As mentioned above, two-photon excitation can be used to cause any kind of optical transition, not just excitation of fluorescence. In fact, two-photon excitation microscopy is extremely useful to produce controlled photochemical reactions in selected regions of a specimen, because its nonlinear nature constrains the two-photon events to the focal volume ($\sim 1 \mu\text{m}^3$). For example, it is possible to produce a desired reaction inside a cell without any activation outside or on the cell surface. The nature of the reaction depends of course on the molecules that have been introduced in the specimen, but photoactivation is commonly used for the uncaging of selected photolabile compounds and photoswitching of fluorescent molecules. In all of these cases, the same reactions could be triggered using confocal microscopy, but not with the same spatial confinement that two-photon excitation provides. Further, most photoactivation processes utilize UV light, so the two-photon effect is also useful for photochemistry as it is for use with UV fluorescence as described above. One elegant example of uncaging using two-photon use is presented in (Judkewitz *et al.*, 2006).

Other kinds of multiphoton microscopy

Multiphoton excitation encompasses a wide range of techniques beyond the most commonly used method of two-photon excitation. Three-photon excitation and second harmonic imaging are two examples. In three-photon excitation, three photons are absorbed simultaneously and each of these provides one-third of the energy needed to reach the excited state. For example, infrared light at 1050 nm can be used to excite UV fluorescence that would normally be excited with 350 nm light. However, the higher order of nonlinearity means even weaker signals (Maiti *et al.*, 1997). Because this

consequently requires higher photon fluxes, the technique has not been used as extensively as the two-photon excitation.

Harmonic generation imaging is fundamentally different: it is a nonlinear optical process, although it is not really a multiphoton one. Second harmonic imaging still requires high photon densities because it relies on high electric fields, so the laser requirements for harmonic imaging is similar to what is used for two-photon excitation. In a fluorescence event, photons are absorbed by the molecules. By contrast, for optical harmonic generation, photons are not absorbed, but rather scattered by the molecules. Scattering is the process by which a photon changes its direction when interacting with particles or molecules. It is particularly strong in biological specimen because of their intrinsic heterogeneity. In a second harmonic generation process, the incoming photons interact with the molecules without undergoing absorption. Instead, they are simultaneously scattered by the molecule, combining to generate a single photon that has double the energy of the incident photons (alternatively, the scattered light has a wavelength that is half of the incident wavelength). Second harmonic generation requires that the molecules in the specimen lack inversion symmetry and that they are spatially ordered. For this reason, second harmonic generation can be used to image ordered structures like collagen fibres or microtubules, with the advantage that it does not require any labelling of the structures (Wang *et al.*, 2010). Third harmonics can also be generated. Third harmonic imaging does not require lack of inversion symmetry in the specimen, and so it can be used to image more disordered molecules (Debarre *et al.*, 2006).

Further reading

There are many detailed review articles covering the different aspects of two-photon excitation, several of which have been cited above. In addition, there are valuable libraries of two-photon absorption spectra for commonly used genetically encoded fluorescent proteins (Drobizhev *et al.*) and endogenous fluorophores (Wang *et al.*, 2010). A significant amount of information on fluorophores and equipment for multiphoton microscopy is also available on the web site of the US National Institutes of Health Resource for Biophysical Imaging (<http://www.drbio.cornell.edu/>).

Conclusion

Multiphoton microscopy today makes full use of a physical phenomenon whose existence was predicted 80 years ago. Thanks to the development of turnkey mode-locked laser systems, multiphoton microscopy is now available for everyone to use without extreme complexity. It produces excitation that is confined to the focal plane, because of the nonlinear nature of the multiphoton interaction with the specimen. This feature results in inherent optical sectioning capability, without the need for a pinhole in front of the detector. As a consequence, photon collection efficiency is greater, and can be further improved through the use of nondescanned detection. Moreover, the confined excitation reduces the photodamage and photobleaching of the specimen. Multiphoton microscopy can image thick samples up to 1 mm, and is often the best technique to perform *in vivo* optical microscopy. It is also useful when UV excitation is necessary, as it uses less toxic infrared light to excite UV-absorbing molecules. We are confident that with the diffusion of this technique in more and more laboratories, and with the design of specific probes that fully utilize the advantages of multiphoton excitation, more experiments will be feasible, including many that we could only dream about just few years ago.

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